Master Thesis

Enhancing Vehicle Interior Action Recognition using Contrastive Self-Supervised Learning with 3D Human Skeleton Representations

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Study Program
M.Sc. Artificial Intelligence and Data Science

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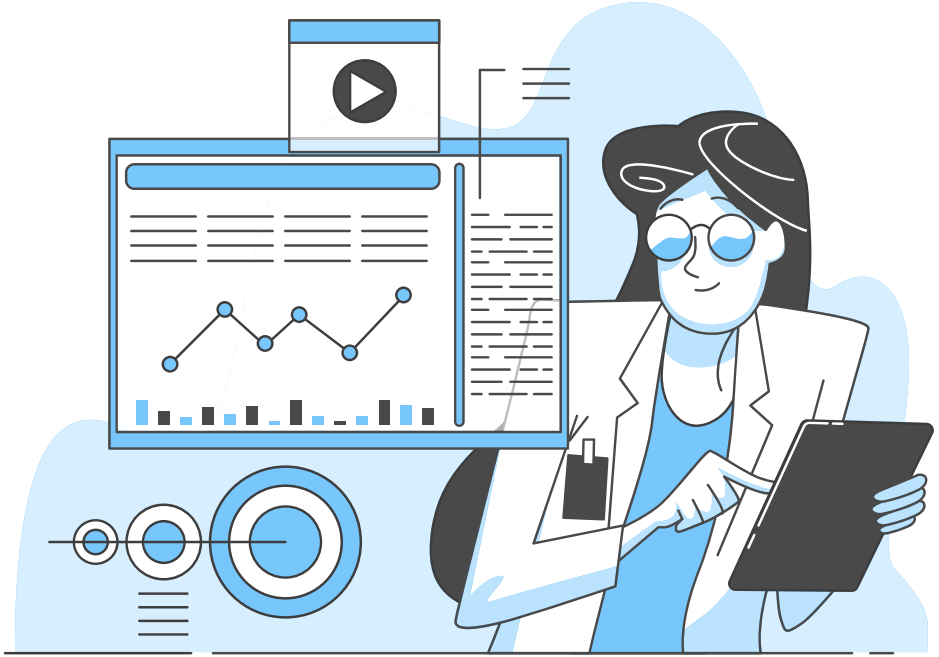
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01

Introduction & Problem Statement

Introduction & Problem Statement



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- According to the WHO, driver distraction is a major contributing factor to road traffic accidents worldwide. Globally, it's estimated that **20-30%** of all road traffic accidents are caused by driver distraction*
- In the EU, it's estimated that driver distraction is a contributing factor in **10-30%** of all road accidents.*
- The rise of **autonomous vehicles** has increased the demand for **action recognition** systems in the car interior.

* (WHO, 2011), * (European Transport Safety Council, 2020).

Introduction & Problem Statement

Problems:

- Traditional **ML** requires annotated datasets: Time-consuming, Costly, Requires expert knowledge
- Generalization to unseen actions, and not only limited labeled data
- RGB, Depth sensors, often come with significant computational costs
- Limited availability of driver action recognition datasets

Introduction & Problem Statement

Solutions:

- **Self-supervised learning** reduces the dependence on manual labeling
- SSL uses larger amounts of unlabeled data
- SSL models tend to learn more general representations
- SSL methods can scale effectively to large datasets and distributed computing environments

- Apply SSL on **3D human skeleton data, where it provides a compact representation of human behavior**

- The utility of the **Drive&Act dataset**

- **What is Action Recognition?**
- **Self-Supervised Learning?**
- **Human Skeleton Data?**

Introduction & Problem Statement

Objective:

1. Develop and evaluate contrastive self-supervised learning approach to recognize human actions. **(label-free, Only 3D Skeleton Data)** – *NTU Dataset*

2. Enhance safety driving by recognizing human actions using SSL with 3D skeleton representations in the vehicle interior. **(label-free, Only 3D Skeleton Data) –** *Drive&Act Dataset*

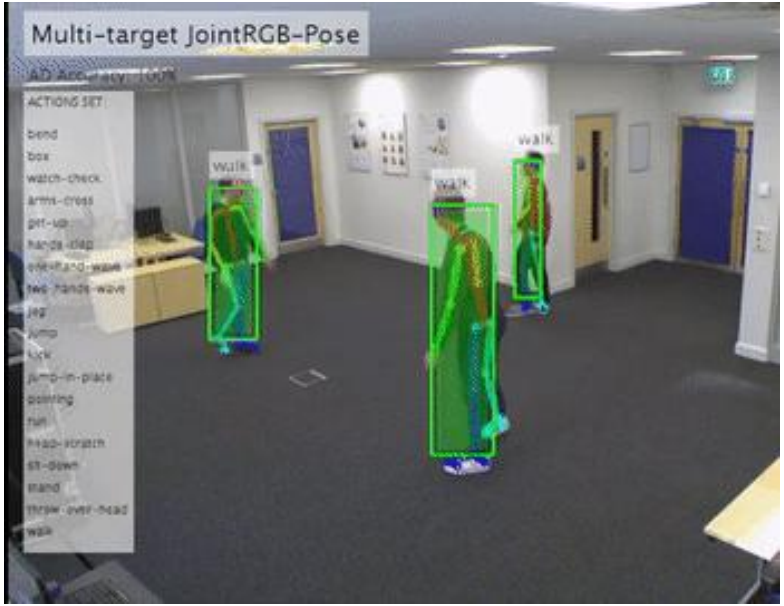
02

Action Recognition & Self-Supervised Learning

Action Recognition

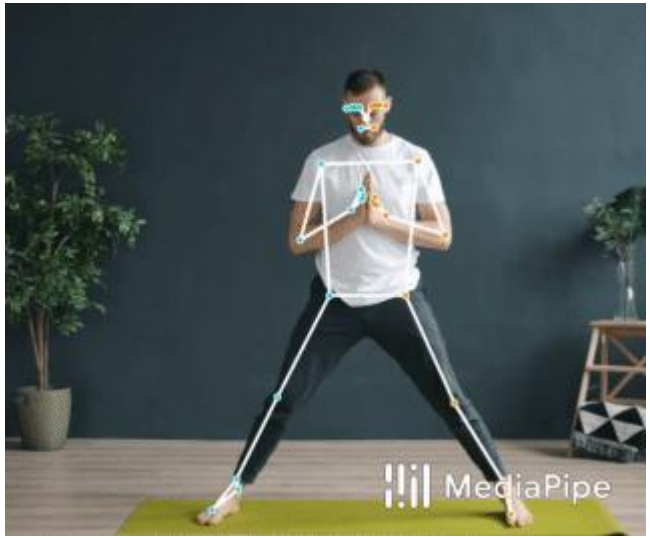
- Action recognition is a field of computer vision using ML/DL techniques to detect and **classify** human actions from video or sensor data in real-time.

- In the context of vehicle safety, action recognition can detect potentially dangerous driver behavior: **texting while driving / looking away from the road / falling asleep at the wheel**.

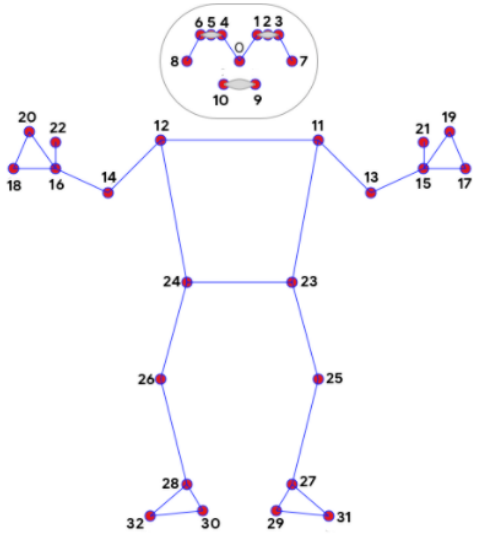


Source: Intellsensingncl NCL Channel

Action Recognition Human Skeleton Based



3D Human Skeleton Data

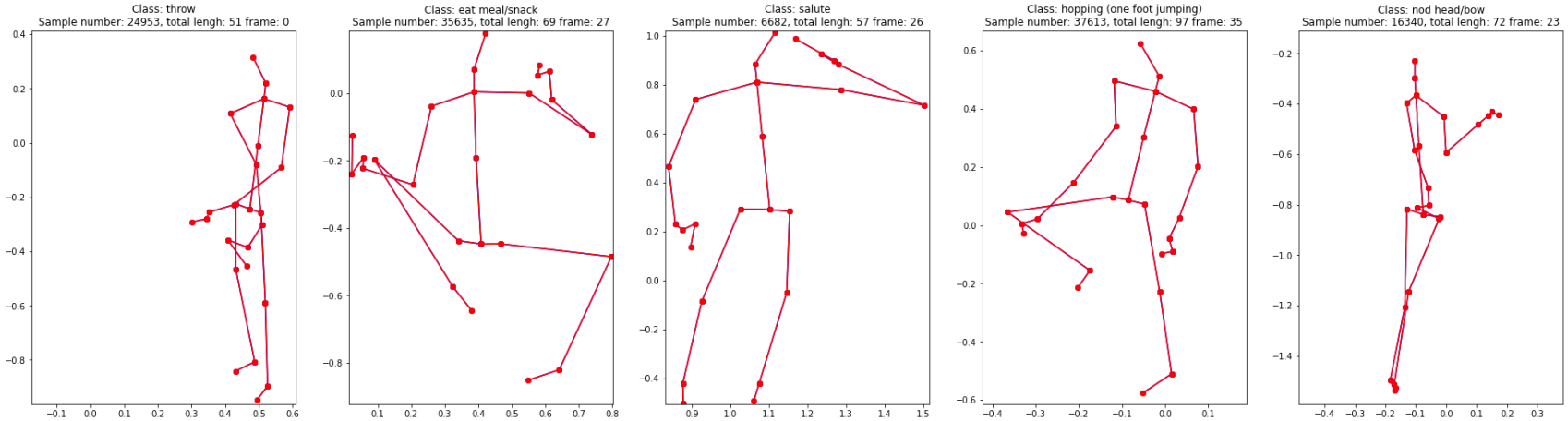


0. nose
1. left_eye_inner
2. left_eye
3. left_eye_outer
4. right_eye_inner
5. right_eye
6. right_eye_outer
7. left_ear
8. right_ear
9. mouth_left
10. mouth_right
11. left_shoulder
12. right_shoulder
13. left_elbow
14. right_elbow
15. left_wrist
16. right_wrist
17. left_pinky
18. right_pinky
19. left_index
20. right_index
21. left_thumb
22. right_thumb
23. left_hip
24. right_hip
25. left_knee
26. right_knee
27. left_ankle
28. right_ankle
29. left_heel
30. right_heel
31. left_foot_index
32. right_foot_index

33 Joints of skeleton data format

Action Recognition Human Skeleton Based

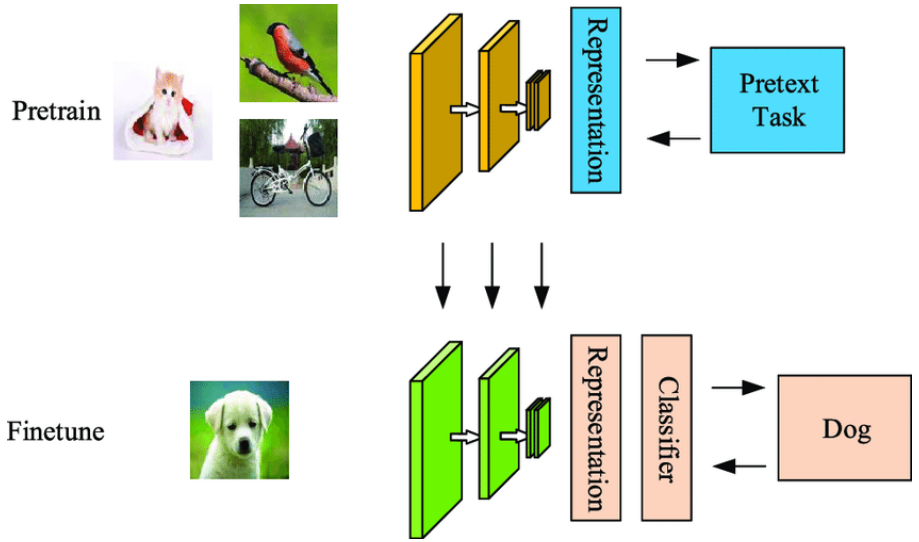
- Action recognition human skeleton-based uses the skeletal structure of the human body to identify specific actions and movements.
- Other modalities exist as RGB Videos, Depth sensors...
- Human skeleton is robust to variations in lighting and background and requires less computational power than other methods.



Different samples of 2D skeletons from NTU Dataset

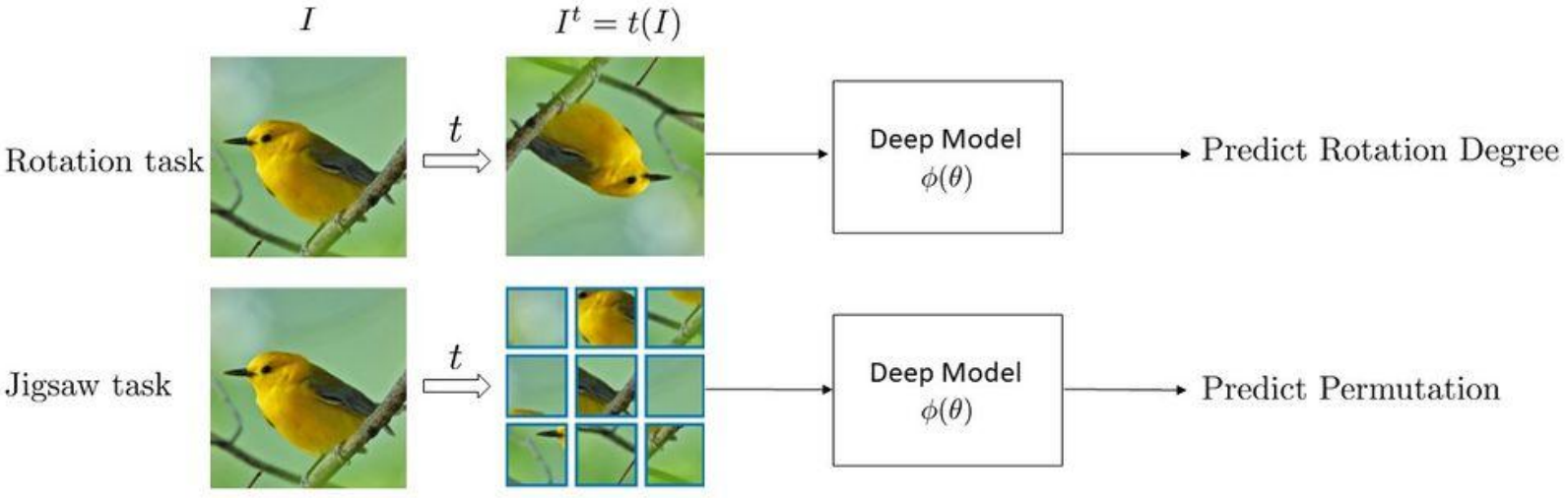
Self-Supervised Learning

- Self-supervised learning is a new type of machine learning where the **algorithm learns from the data itself**, without relying on labeled data from humans.
- Self-supervised learning has become popular in computer vision because it allows for large-scale datasets without manual annotation and can improve model generalization to new data.



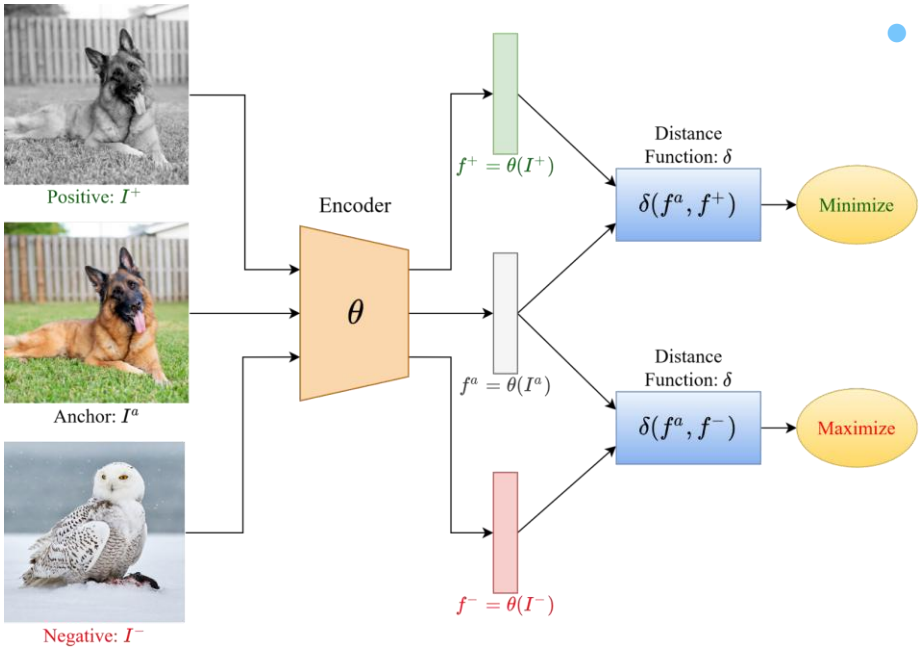
Self-Supervised Learning Diagram

SSL: Pretext Task



Examples of SSL pretext tasks. (a) Rotation task, (b) Jigsaw puzzle task

SSL: Contrastive learning



Contrastive Learning Diagram

- **Contrastive learning** is a type SSL in computer vision that learns representations by contrasting positive and negative pairs of data.

Contrastive learning has shown promising results in image recognition, object detection, and other computer vision applications and is an active area of research in the field.

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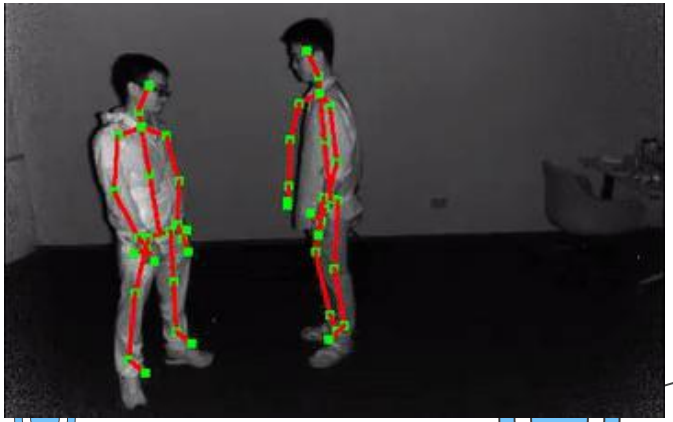
Datasets: NTU RGB / Drive&Act

NTU RGB+D Dataset

- NTU RGB+D dataset, is a large-scale benchmark for 3D human activity understanding.
- Contains over **114 thousand video** samples and **8 million frames**, collected from **106 distinct subjects**, and includes **60 different action classes**.
- The Dataset was captured by 3 different modalities: RGB, depth, infrared, and skeleton data.



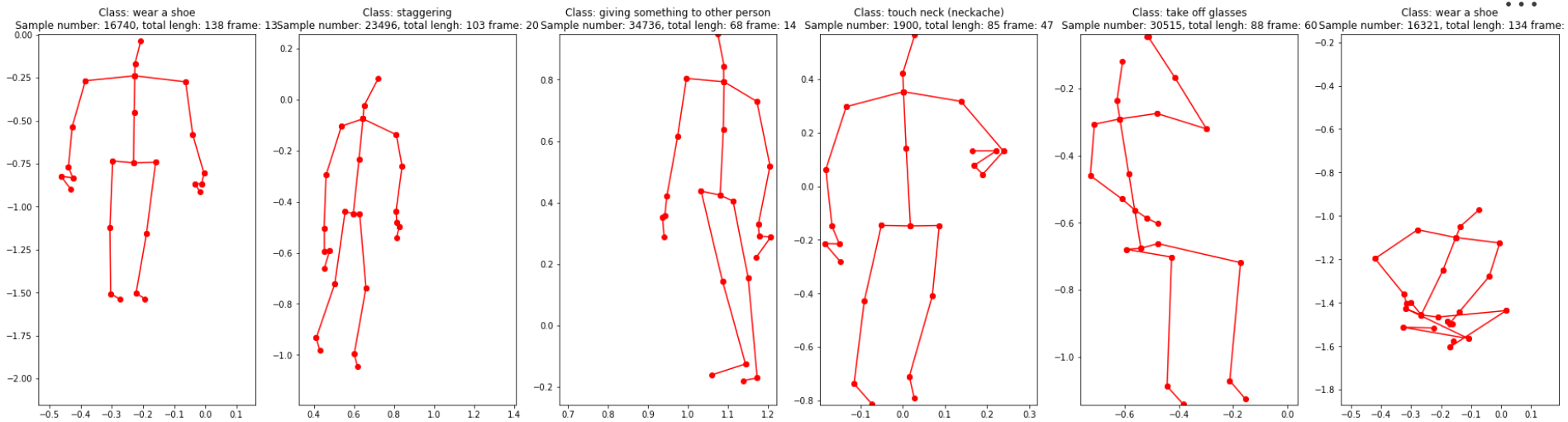
RGB with Skeleton Data



IR with Skeleton Data

NTU RGB+D Dataset

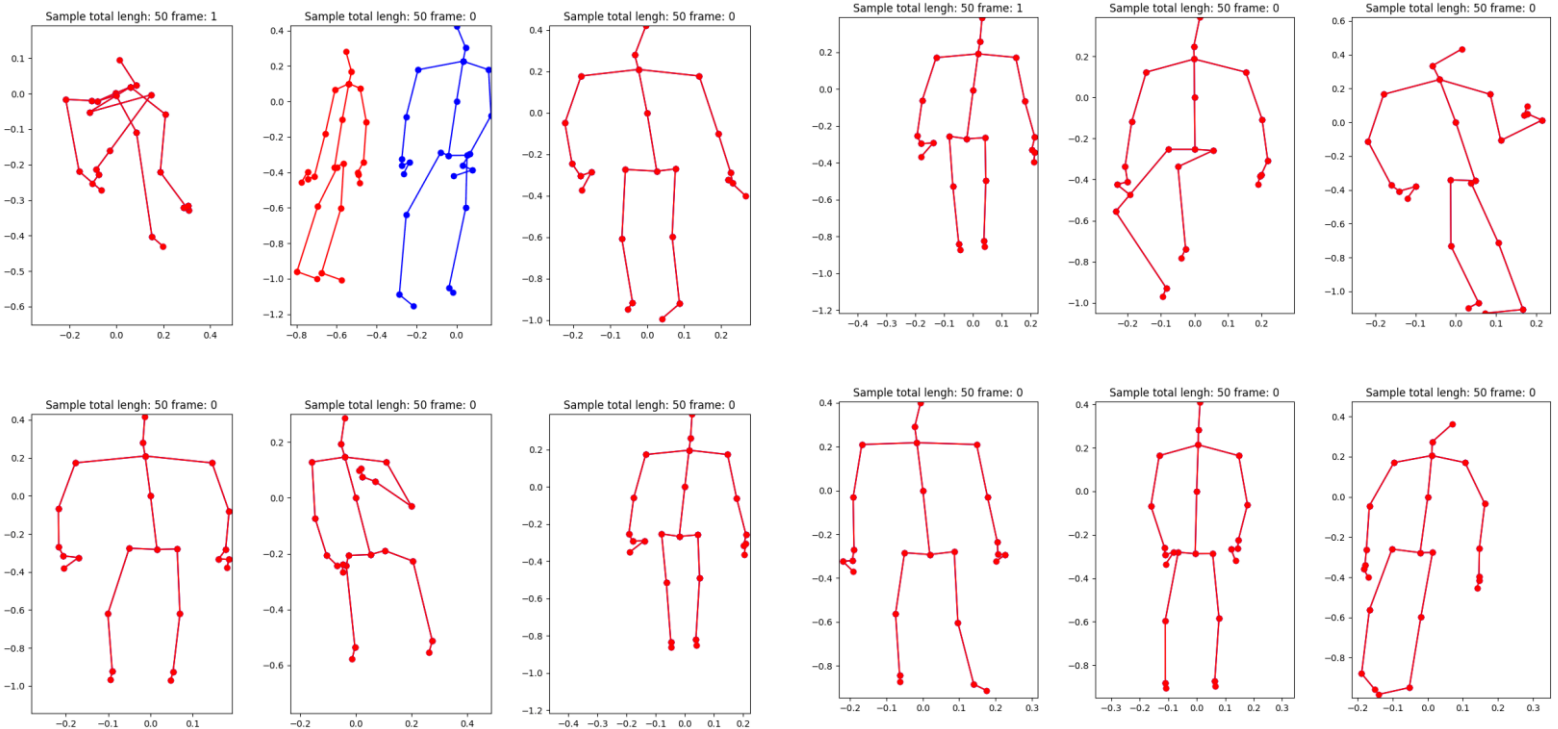
- **Examples of action labels:** drink water, eat meal/snack, brushing teeth, brushing hair, drop, pickup, throw, sitting down, standing up.



Different samples of 2D skeletons from NTU 60 Dataset

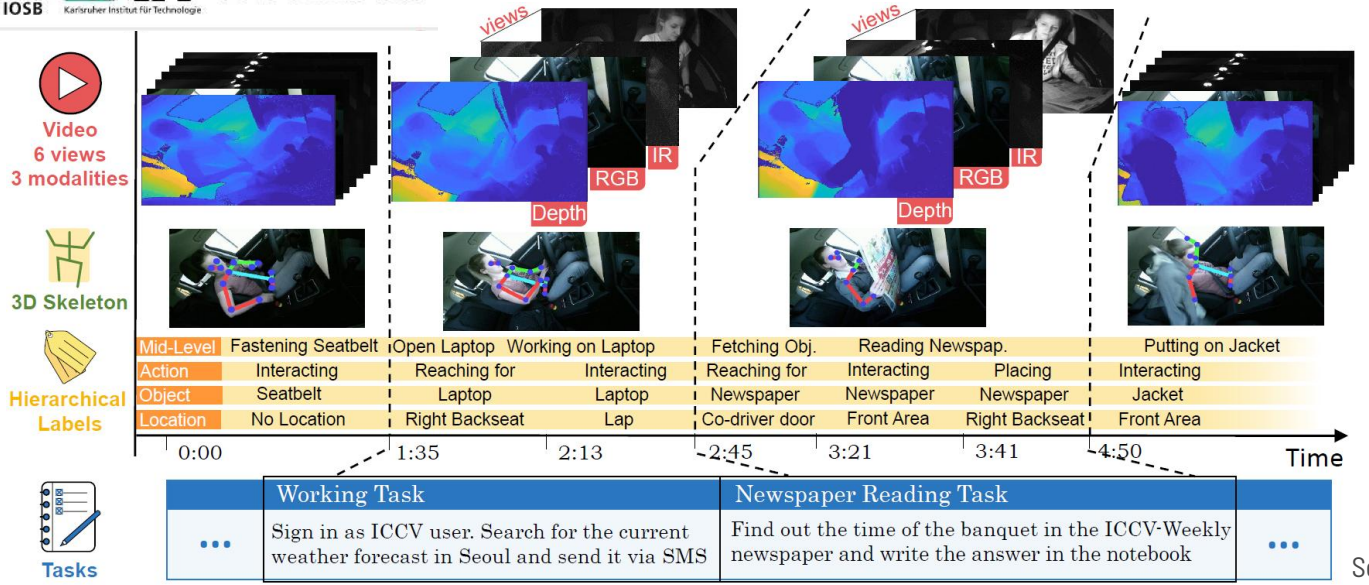
NTU RGB+D Dataset

- **Examples of action labels**: drink water, eat meal/snack, brushing teeth, brushing hair, drop, pickup, throw, sitting down, standing up.



Multiple frames from a sample of 2D skeletons from NTU RGB+D dataset

Drive&Act Dataset

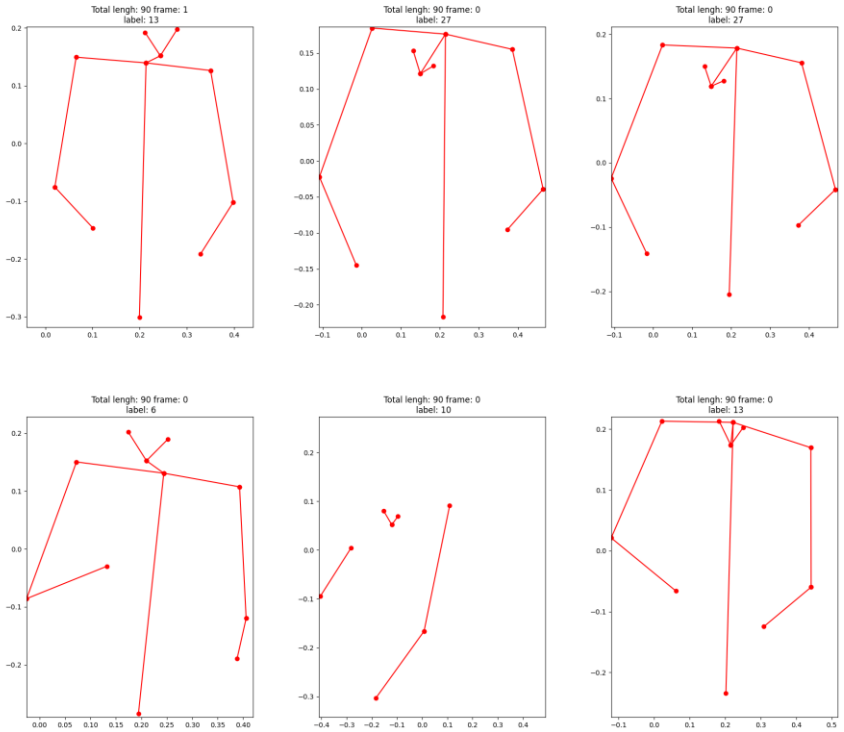


Source: driveandact.com

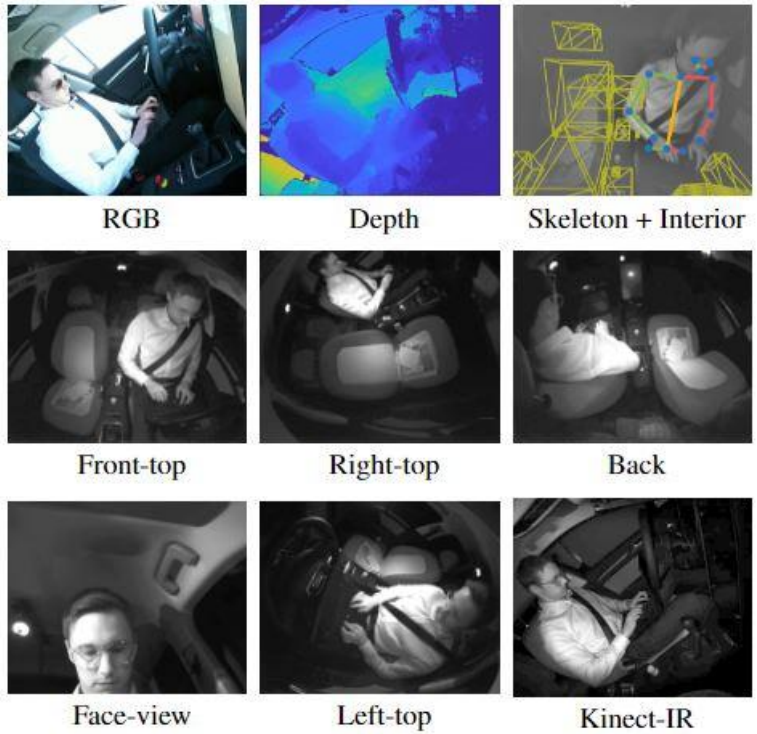
- The Drive&Act dataset is a state-of-the-art multi-modal benchmark for action recognition in automated vehicles.
- **12h of video** data in **29 long sequences**
- Multi-modal videos: NIR, Depth, RGB data.
- Contains 34 classes

Drive&Act Dataset

Examples of action labels: standing up, sitting, reaching for, interacting, putting on



Visualization of skeleton sequence data from the Drive&Act dataset with 11 joints.



Different camera views and modalities from the Drive&Act dataset.

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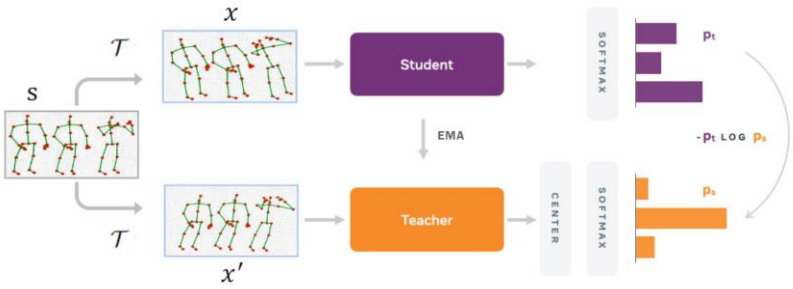
Learning Approaches

Learning Approaches

First Approach:

Encoder: **STTFormer**

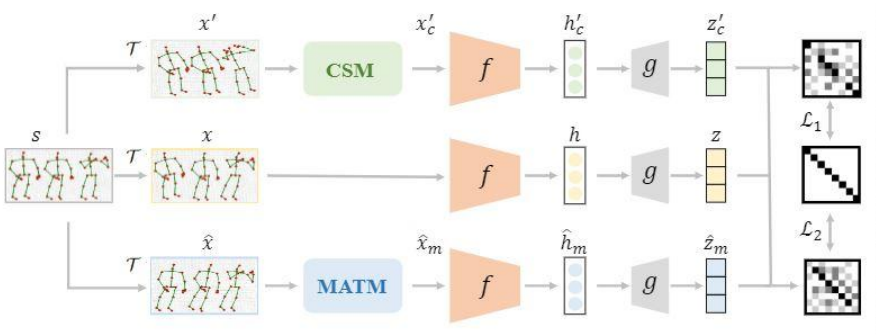
Learning Strategy: **DINO**



Second Approach:

Encoder: **ST-GCN**

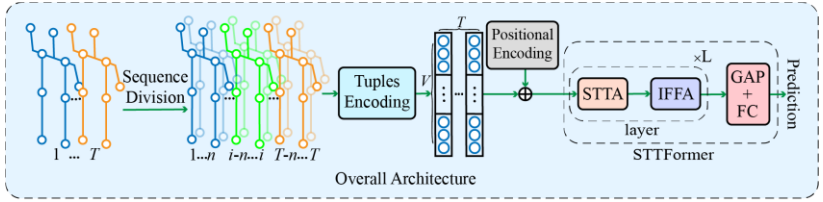
Learning Strategy: **PSTL**



Learning Approaches

First Approach:

Architecture: **STTFormer**

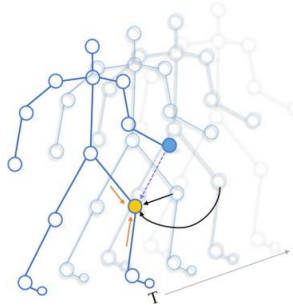


Spatio-Temporal Tuples Transformer for Skeleton-Based Action Recognition

- Novel method for skeleton-based action recognition that captures the dependencies between joints and achieves better performance on large-scale datasets.
- It consists of two main blocks: the Spatio-Temporal Tuples Attention (STTA) and Inter-Frame Feature Aggregation (IFFA).

Second Approach:

Architecture: **ST-GCN**



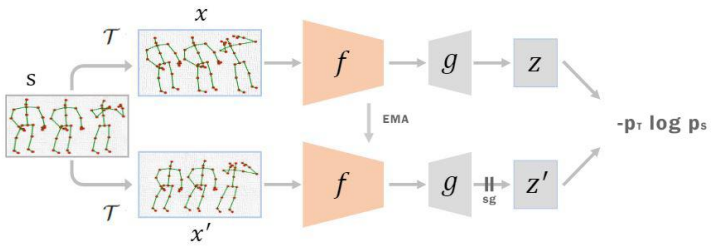
The Spatial Temporal Graph Convolutional Networks

- The motivation behind using graph convolution networks arises from skeletons being inherently represented in graphs, where each node corresponds to a joint of the human body.
- It provides a graph-based alternative to standard 2D convolutions and enables the model to effectively handle skeletal data for action recognition tasks.

Learning Approaches

First Approach:

Learning Strategy: **DINO**

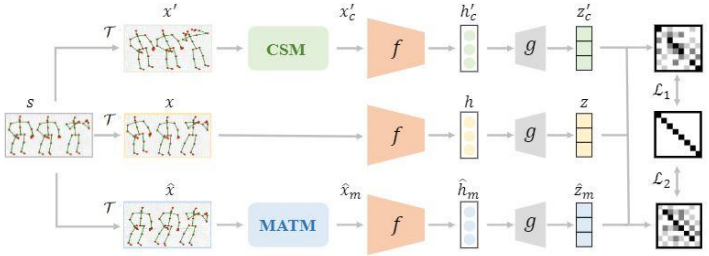


Self-*DI*stillation with *NO* labels

- The model processes five different random transformations of an input.
- These transformations are passed through the student network, and the teacher network.
- The core of the DINO framework lies in the self-distillation process.

Second Approach:

Learning Strategy: **PSTL**



Partial Spatio-Temporal Learning

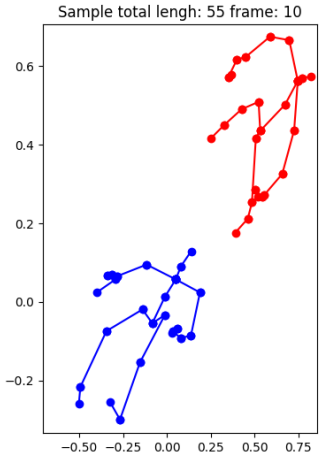
- PSTL adopts a triple-stream architecture.
- It applies Central Spatial Masking (CSM) on the spatial masking stream and Motion Attention Temporal Masking (MATM) on the temporal masking stream
- Extra anchor stream is included to retain the original semantic information.

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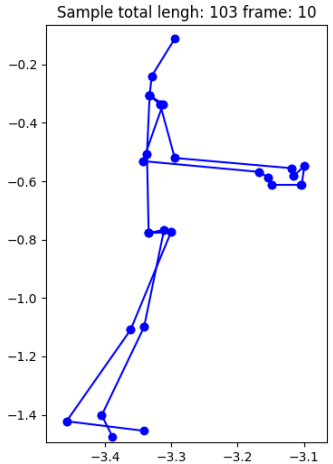
Experiments & Results

Data Preparation and Processing

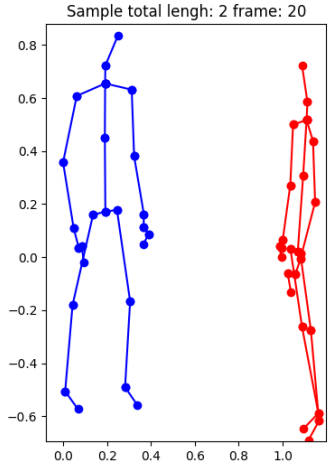
- Build datasets pipeline:
  - Visualizations
  - Normalizations
  - Augmentations: Rotation, Cropping, Spatial Flip, Noise, Masking...



Apply rotation to a sample

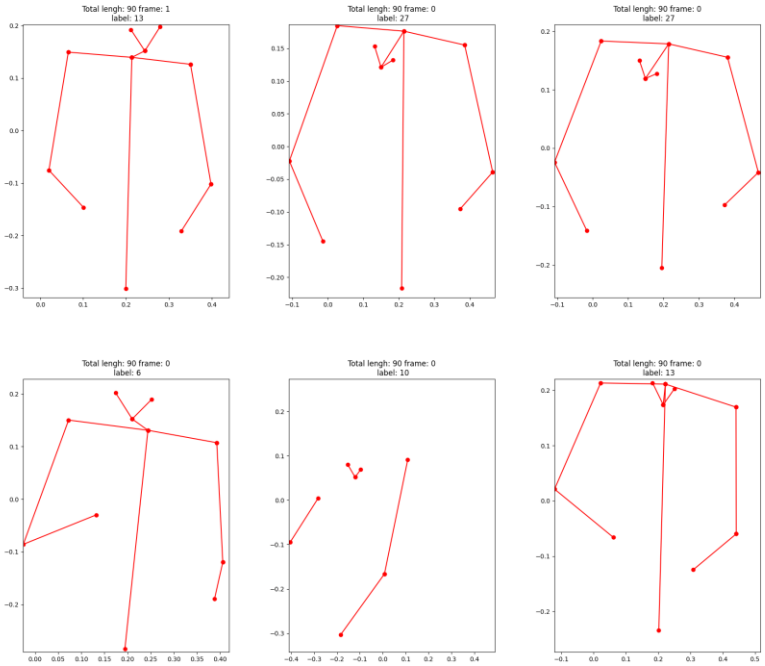


Visualizing data on x,z axis

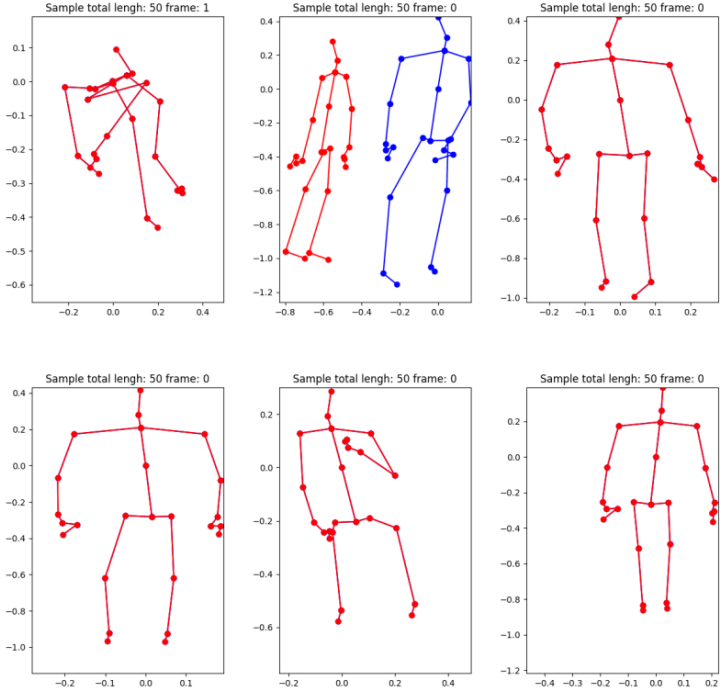


Visualizing multiple subject from a sample

Data Preparation and Processing



Visualization of Drive&Act as animation

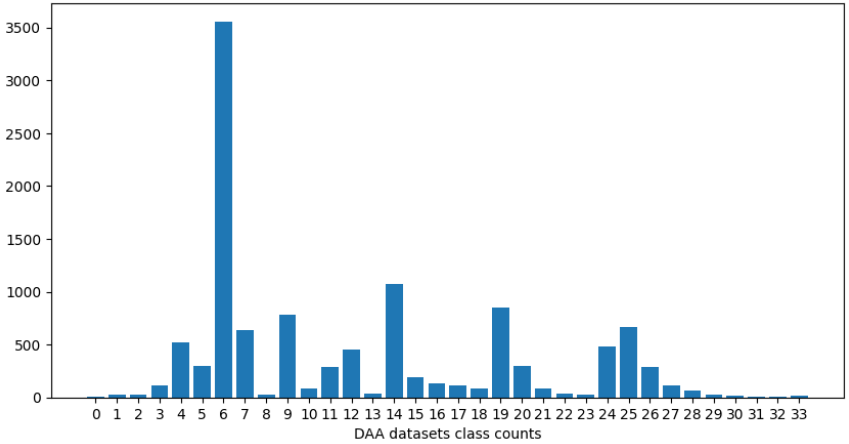


Visualization of NTU RGB+D as animation

Data Preparation and Processing

Drive&Act Challenges:

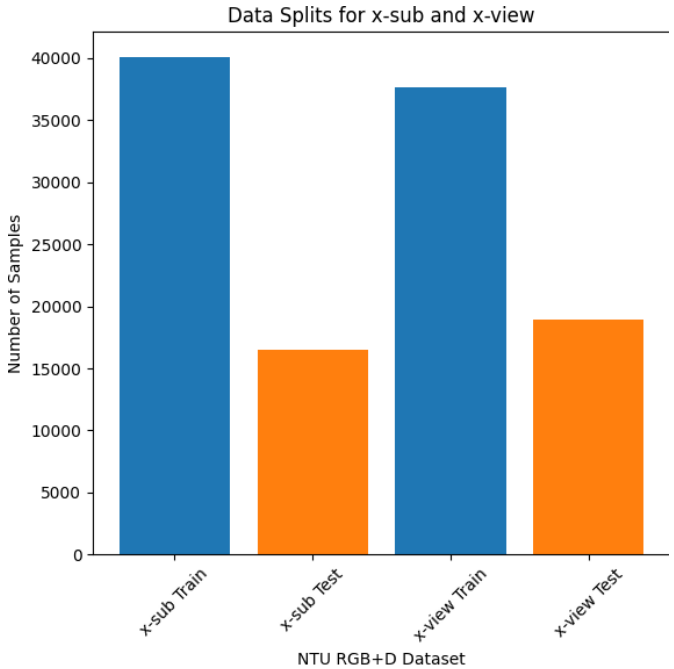
- Number of errors in the annotation, including missing joints, shaded joints, and incorrect joint coordinates.
- Not all joints were consistently visible in the camera frame, particularly in the lower body region.
- Actions exhibited similarity in terms of joint movements, making it challenging for the model to discriminate between action classes effectively.
- Dataset suffered from a significant class imbalance, with the class "sitting" being the most dominant among the 34 classes with 3581 samples.



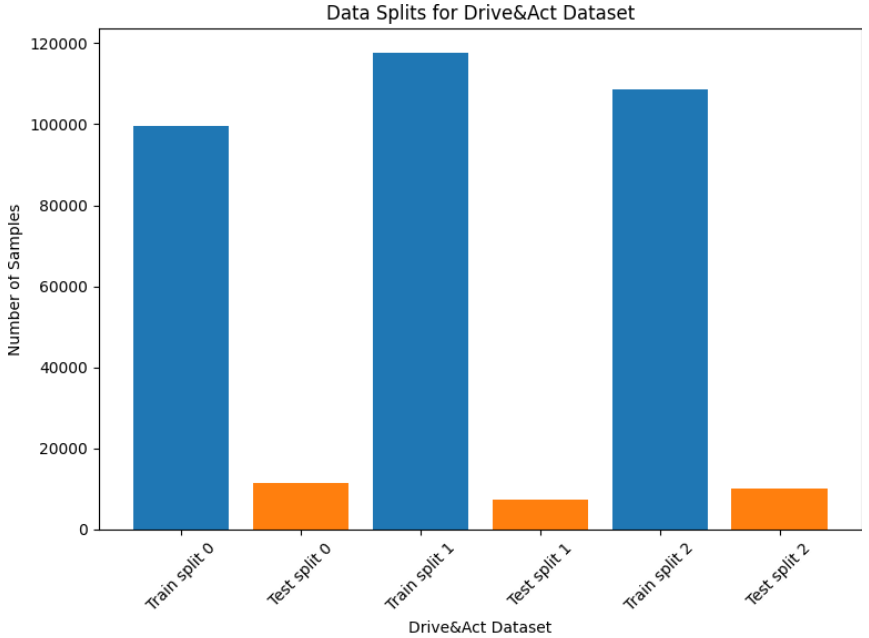
Bar plot describing the class distribution of the Drive&Act dataset.

Data Preparation and Processing

NTU RGB+D Splits:



Drive&Act Splits:



05

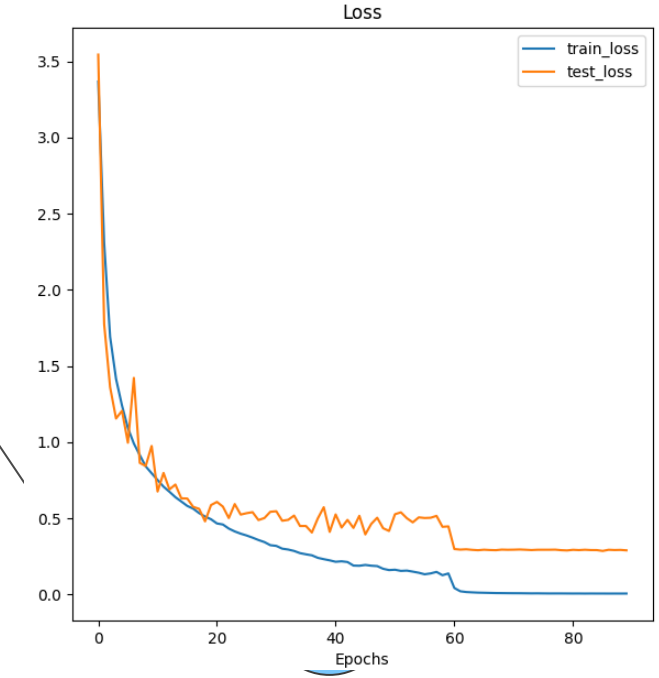
Results

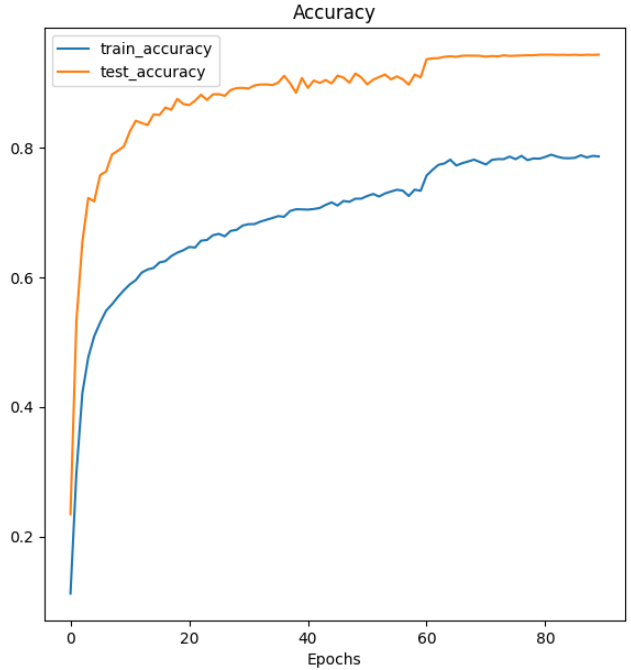
Supervised Learning Results

- Train the model on 90 epochs.
- Use of adaptive learning rate
- Train on 4 GPUs (NVIDIA RTX A6000) for 9 hours.

Table 5.1: Supervised Learning Results on NTU RGB+D Dataset

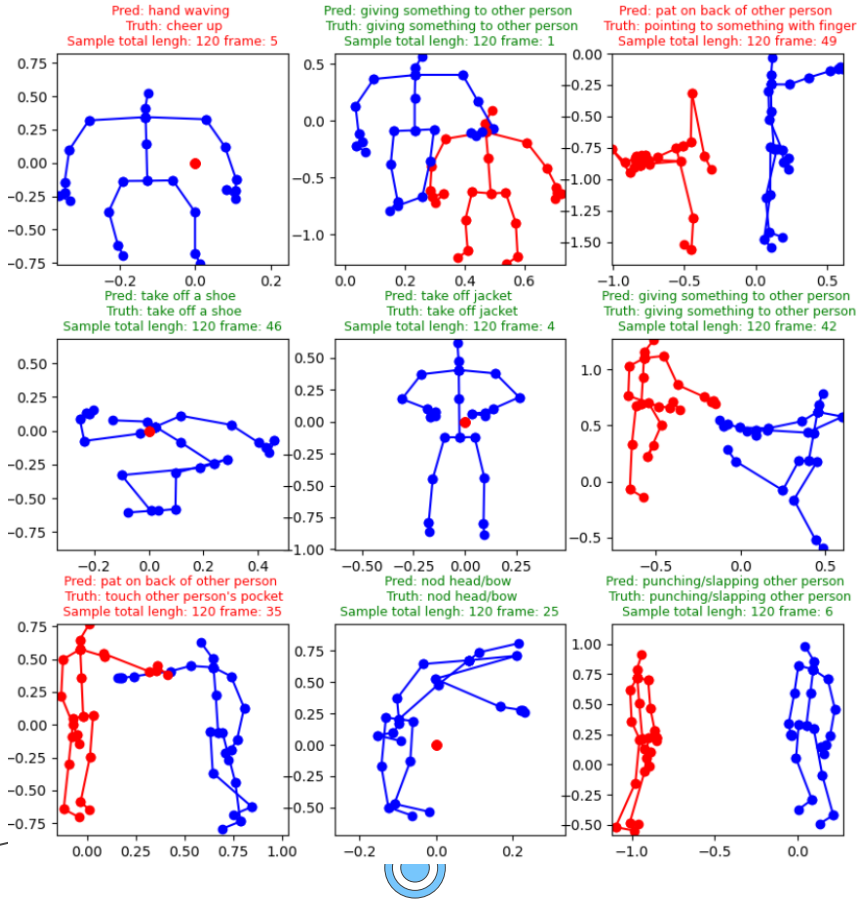
Encoder	NTU RGB+D (X-Sub)	NTU RGB+D (X-View)
ST-GCN	81.5%	88.3%
STTFormer	84.3%	94.3%





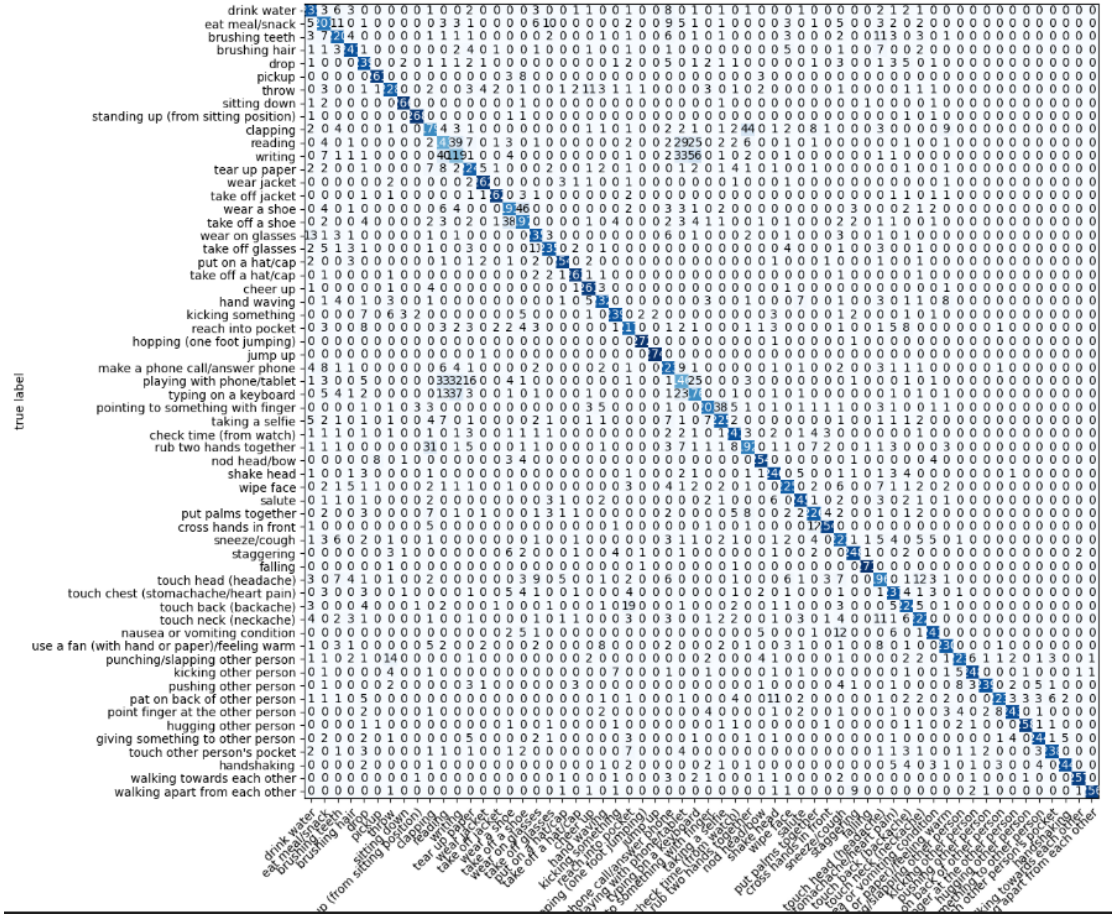
STTFormer learning curve shows the loss function on the left and the accuracy metric on the right over the epochs

Supervised Learning Results



Plot showing STTFormer false and true predictions on NTU RGB+D dataset

Supervised Learning Results



Plot showing STTFormer Confusion Matrix for NTU RGB+D Dataset

Self-Supervised Learning Results

SSL Evaluation protocol:

- k-Nearest Neighbors
- Linear Evaluation
- Finetuning
- Cross-Evaluation

Table 5.3: SSL Results on NTU RGB+D Dataset with k-NN evaluation

Approach	Evaluation Type	Accuracy (%)
DINO & STTFormer	k-NN (X-sub)	23.02
DINO & STTFormer	k-NN (X-view)	30.04
PSTL & ST-GCN	k-NN (X-sub)	58.65
PSTL & ST-GCN	k-NN (X-view)	66.06

Table 5.5: Results of PSTL & ST-GCN Approach

Dataset	Evaluation Type	Accuracy (%)
NTU RGB+D	Linear Evaluation (X-Sub)	76.76
NTU RGB+D	Linear Evaluation (X-View)	82.56
NTU RGB+D	Finetune (X-Sub)	84.2
NTU RGB+D	Finetune (X-View)	91.6
Drive&Act	Linear Evaluation	14.6
Cross-Evaluation	NTU RGB+D to Drive&Act	3.91

Self-Supervised Learning Results

Ablation Studies on Drive&Act with PSTL & ST-GCN approach:

Table 5.6: Joint Variation Experiment:

Dataset	Number of Joints	k-NN Evaluation Accuracy
Drive&Act	11 joints	5.37%
Drive&Act	8 joints	4.3%

Table 5.7: Sequence Length Experiment:

Dataset	Number of Frames	k-NN Evaluation Accuracy
Drive&Act	90 frames	5.37%
Drive&Act	50 frames	3.98%

Linear evaluation results of the state-of-the-art on NTU RGB+D dataset. * our implementation of the approach.

Method	NTU-60 (%)	
	xsub	xview
MS2L (ACM MM 20) [57]	52.6	-
P&C (CVPR 20) [56]	50.7	76.3
AS-CAL (Inf Sci 21) [58]	58.5	64.8
AimCLR (AAAI 22) [61]	74.3	79.7
PSTL & ST-GCN	77.3	81.8
PSTL & ST-GCN *	76.8	82.6

06

Conclusion

Conclusion

- Two approaches investigated: DINO with STTFormer and PSTL with ST-GCN. **PSTL with ST-GCN outperformed**, also demonstrating a large gap between NTU and Drive&Act.
- **Challenges in applying DINO with STTFormer** due to DINO developed on image data with extensive information, in contrast with the compressed nature of skeleton data. Suggested exploration of alternative encoders like ST-GCN within DINO framework.
- **Drive&Act** dataset exploration faced annotation errors, imbalanced classes. Future work is essential for improved skeleton modality techniques and quality datasets.
- Built tools and pipelines (e.g. augmentations) for processing **3D skeleton data**.
- Analyzing and evaluating skeleton data, constructing complex models, employing various evaluation methodologies **(k-NN, Linear evaluation, finetuning, cross-evaluation),** and addressing challenges posed by different datasets.

Thank You for Your Attention!

Fraunhofer IOSB

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